

	B.M.S. COLLEGE OF ENGINEERING, BANGALORE-19		
Department of Computer Science and Engineering			
INTERNALS-1			
Course Code: 23CS4PCADA	Course Title: Analysis and Design of Algorithms		
Semester: 4	Maximum Marks: 40	Date: 08-04-2026	
Faculty Handling the Course:	Dr.KSS, PA, MS, Dr.JSN, Dr.BSR, Dr.NM, Dr.HSG, Dr.ST, AAS, DK, MNM, RP, HSM		
<i>Instructions: Internal choice provided in Part C.</i>			

PART-A

Total 5 Marks (No Choice)

No.	Question	Marks	CO-PO	Bloom's Level
1	Solve the following recurrence relation using backward substitution method. i. $T(n) = T(n/2) + n$ for $n > 1$, $T(1) = 1$ ii. $X(n) = 2 * X(n-1) + 1$ for $n > 1$ and $X(0) = 0$	5	1-2	L4

PART-B

Total 15 Marks (No Choice)

No.	Question	Marks	CO-PO	Bloom's Level
2a)	A list contains employee records sorted by department. Now, it needs to be sorted by salary without disturbing department order for equal salaries. Suggest the best algorithm for this and justify. Also provide the algorithm and analyze its time efficiency.	5	1-2	L4
2b)	Consider the algorithm given below and answer the questions that follow: ALGORITHM Enigma ($A[0..n-1, 0..n-1]$) //Input: A matrix $A[0..n-1, 0..n-1]$ of real numbers for $i \leftarrow 0$ to $n-2$ do for $j \leftarrow i+1$ to $n-1$ do if $A[i, j] \neq A[j, i]$ return false return true i) What does this algorithm compute? ii) What is its basic operation? iii) How many times is the basic operation executed? iv) What is the order of growth of this algorithm?	5	1-2	L4
2c)	Apply Divide and Conquer technique to multiply the following two long integers: 1234 and 5678. Justify the significance of divide and conquer technique over the traditional method.	5	2-1	L3

PART C

Total 20 Marks

No.	Question	Marks	CO-PO	Bloom's Level																		
3a)	A university maintains a large database of student records, where certain attributes (like grades, attendance status, and course outcomes) are stored as symbols. Due to increasing data volume, the university plans to compress student data for efficient storage and faster retrieval. Design an efficient encoding algorithm to compress the student data. Given the following symbols and their frequencies: <table border="1"> <thead> <tr> <th>Symbol</th><th>A</th><th>B</th><th>C</th><th>D</th><th>E</th></tr> </thead> <tbody> <tr> <td>Description</td><td>Outstanding</td><td>Distinction</td><td>First Class</td><td>Second Class</td><td>Pass</td></tr> <tr> <td>Frequency</td><td>40</td><td>10</td><td>20</td><td>15</td><td>15</td></tr> </tbody> </table> i) Encode ABACABAD and ii) Decode 100010111001010 using the generated code.	Symbol	A	B	C	D	E	Description	Outstanding	Distinction	First Class	Second Class	Pass	Frequency	40	10	20	15	15	10	2-1	L3
Symbol	A	B	C	D	E																	
Description	Outstanding	Distinction	First Class	Second Class	Pass																	
Frequency	40	10	20	15	15																	
OR																						
3b)	A hospital needs to send an ambulance to reach patients across different locations. The travel time between locations due to road conditions and traffic are given below. Apply an appropriate algorithm to determine the shortest time required for the ambulance to reach all locations from the hospital. Also analyze the algorithm for its time efficiency.	10	2-1	L3																		
4a)	Show how the following array is sorted using Quick Sort. Also, write the quick sort algorithm. 54, 26, 93, 17, 77, 31, 44, 55, 20	10	2-1	L3																		
OR																						
4b)	Find the MST for the following Graph using Kruskal's Algorithm and also write the Kruskal's Algorithm.	10	2-1	L3																		

ALL THE VERY BEST